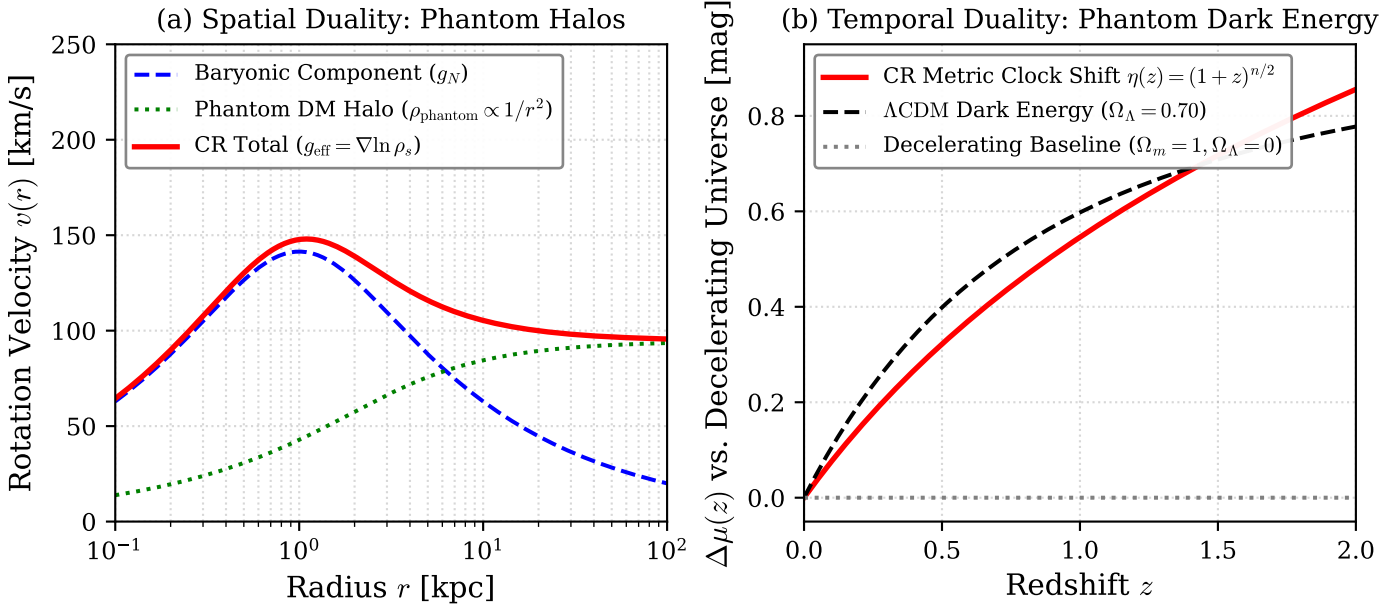
(a) Spatial Duality: Phantom Halos
Baryonic Component (g_N)
Phantom DM Halo ($\rho_{\text{phantom}} \propto 1/r^2$)
CR Total ($g_{\text{eff}} = \nabla \ln \rho_s$)
Rotation Velocity $v(r)$ [km/s]
Radius r [kpc]
250
200
150
100
50
0
10^{-1}
10^{0}
10^{1}
10^{2}
(b) Temporal Duality: Phantom Dark Energy
CR Metric Clock Shift $\eta(z) = (1+z)^{n/2}$
ΛCDM Dark Energy ($\Omega_\Lambda = 0.70$)
Decelerating Baseline ($\Omega_m = 1, \Omega_\Lambda = 0$)
$\Delta\mu(z)$ vs. Decelerating Universe [mag]
Redshift z
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0